CO2 · ASSESSMENT TOOL - I N. SRIHARI

ANALYTICAL PROBLEM SOLVING 192572127

29/50

1. a) Purpose of Grey Level Transformation

* Enhance image contrast
* Highlight important features
* Improve brightness
* Prepare images for further process.

2

b). $S = 255 - r$.

Orginal pixel (r)

50, 100, 150, 200

1) $255 - 50 = 205$

2) $255 - 100 = 155$

3) $255 - 150 = 105$

4) $255 - 200 = 55$.

Transformed pixels : 205, 155, 105, 55

2. Histogram Equalization

Given

Intensity	Frequency
0	4
1	6
2	10
3	6.

Total number of pixels :
N = 4+6+10+6 = 26

a) * Enhances low contrast images
* Improves visibity of details
* utilizes the full intensity range.

pdf

b) .Compute CDF

$\frac{\text{Frequenc}}{\text{Total no of freque}} = 4/26 = 0.154$

10/26 = 0.385

20/26 = 0.769

26/26 = 1

c). S = round ((L-1) × CDF)

L = 4 L-1 = 3.

round (3× 0.154) = 0

round (3×0.385) = 1

round (3×0.769) = 2

round (3×1) = 3

* It removes noise smoothly.
* Gives higher weight to the center pixel
* Produces less blurring than an average filter.

N SRIHARI

b). Gaussian kernel

kernel

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image : $\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$

$(10\times1)+(20\times2)+(10\times1)+(20\times2)+(40\times4)+(20\times2)+(10\times1)+(20\times2)+(10\times1)$

$= 10+40+10+20+160+40+10+40+10$

$= 360$

$\frac{360}{16} = 22.5$ Rounded value : 23

6

Purpose of Edge Detection

* object detection
* Image segmentation
* shape recognition
* feature extraction

2

b) Horizontal Sobel mask.

mask $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

Image $\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$

Calculation

$(-1\times10)+(-2\times10)+(-1\times10) = -40.$

middle row

0

Bottom row.

$(1\times 30)+(2\times 30)+(1\times 30) = 120$

Gradient

$G_x = -40 + 120 = 80$

Result: $\boxed{G_x = 80}$

5 a) Contrast streching

* Enhances image visibility.
* makes dark regions darker and bright region brighter
* improves feature detection

uses the full available intensity range

b) $s = \dfrac{(r - r_{min})(255)}{r_{max} - r_{min}}$ $r_{min} = 50$ $r_{max} = 150$.

$s = \dfrac{(r-50)\times 255}{100}$

$r=50$ $s = \dfrac{(50-50)\times 255}{100} = 0$

$r=100$ $s = \dfrac{(100-50)\times 255}{100} = 127.5 \simeq 128$

$r=150$ $s = \dfrac{(150-50)\times 255}{100} = 255$

r	s
50	0
100	128
150	255